Optimal Load Balancing of Cooperative UAV–UGV Parcel Pickup to Minimize Completion Time

Soobin Yoon and Kisong Lee

***Abstract*— In this study, we investigate an optimal load balancing of cooperative parcel pickup between an unmanned aerial vehicle (UAV) and an unmanned ground vehicle (UGV). By considering practical aspects, such as the movement characteristics of each vehicle and the avoidance of no-fly zones for the UAV, we aim to optimize the three-dimensional trajectories and pickup strategies of the two vehicles to identify the shortest route to minimize pickup completion time. To deal with the nonconvex optimization problem, we employ a successive convex approximation to convert the original problem into a convex form for optimization variables, and we also use a penalty convex-concave procedure to retain the binary natures of the control parameters that are required to design the pickup strategy. We also propose a two-stage iterative algorithm based on interior-point methods to solve the relaxed convex problem to find suboptimal solutions. The simulation results confirm that the proposed scheme can successfully allow the UAV and UGV to follow the shortest effective path and thus improve pickup completion time through load balancing, while outperforming the baseline schemes under various scenarios.**

***Index Terms*— UAV–UGV cooperation, load balancing, 3D trajectory, pickup design, nonconvex optimization.**

I. Introduction

Recent years have seen significant advancements in the development and production of cost-effective unmanned aerial vehicles (UAVs), which has led to the increasing versatility of UAVs across a wide range of applications, including military, surveillance, and delivery services [1]. Most notably, Amazon launched a UAV delivery service in the UK in December 2016 that allows parcels weighing up to five pounds to be delivered within 30 minutes of ordering [2]. Accordingly, there have been various studies aiming to identify the optimal strategy for UAV-enabled parcel delivery [3], [4], [5], [6], [7]. In [3], a collision-free path planning algorithm was proposed to find the shortest path on a dynamic graph. In [4], an Internet of Things-based UAV delivery system was developed to provide UAVs with the ability to autonomously fly in mixed indoor and outdoor environments. Moreover, the dynamic meal-order task scheduling problem was explored to minimize total tardiness in a UAV-enabled on-demand meal delivery system [5]. The authors of [6] jointly optimized the trajectory of the UAV and parcel pickup strategy with the aim of maximizing total item value while avoiding no-fly zones (NFZs). In [7], a simulated-annealing-based two-phase optimization approach was designed to solve the scheduling and routing problems for UAV package pickup and delivery systems.

Along with UAVs, unmanned ground vehicles (UGVs) which can move and operate autonomously have been leveraged for delivery applications [8], [9]. In [8], the authors discussed the application of UGVs as delivery robots and developed autonomous UGV delivery systems. In [9], a dynamic algorithm was proposed that allowed a UGV to continuously optimize its route while performing the same task repeatedly. There has also been some recent research examining cooperation between UAVs and UGVs by combining their respective strengths [10], [11], [12], [13], [14], [15]. In [10], a computer vision-based framework was presented for a cooperative UAV–UGV system for path planning. In [11], an obstacle avoidance algorithm was designed that could guide UAVs and UGVs in path-following tasks. The authors of [12] formulated a model involving cooperative UAV–UGV-enabled emergency resource delivery and suggested a three-stage method to solve the nested vehicle routing problem. In [13], the circular paths for UAVs and UGVs were optimized to minimize travel time while achieving complete coverage for surveillance tasks. Finally, a low-complexity parcel delivery scheduling policy was proposed for a UAV-vehicle collaboration system in [14], and collaborative pickup and delivery tasks between trucks and UAVs were optimized to minimize operational costs in [15].

Although existing studies have investigated cooperative strategies for UAV–UGV delivery [10], [11], [12], [13], they have not considered the actual movement characteristics of each vehicle that may occur in real-world scenarios. With this background, we design an optimal load balancing for cooperative UAV–UGV parcel pickup that can be applied in practice in real-world scenarios. The specific contributions of our study can be summarized as follows:

- Taking into account the distinct movement characteristics of each vehicle, such that the UAV can fly freely in three-dimensional (3D) space but cannot access NFZs whereas the UGV can only move in either the horizontal or vertical direction at any given moment, we design a mathematical model for a cooperative parcel pickup scenario. Based on this model, we aim to jointly optimize the UGV movement direction indicator, pickup indicator, 3D trajectory, and lengths of time slots for the two vehicles to identify the shortest route to minimize pickup completion time, which has not been done to this point.
- To approximate suboptimal solutions from the formulated nonconvex problem, we employ successive convex approximation (SCA) to convert the original constraints into convex sets and leverage interior-point methods to solve the transformed convex problem iteratively. We also use the penalty convex-concave procedure (PCCP) to maintain the binary characteristics of the optimization variables concerning the pickup strategy, although they are relaxed to have continuous values to solve the problem. Finally, we propose a two-stage iterative algorithm to derive the UAV–UGV cooperative strategy.
- The results of extensive simulations conducted under a variety of scenarios confirm that the proposed scheme not only minimizes the pickup completion time but also forms the shortest route without violating NFZs through two-stage optimization. It also outperforms the baseline schemes in terms of completion time

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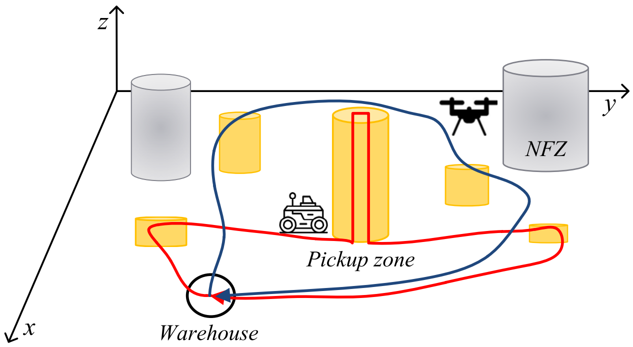


Fig. 1. System model of a cooperative parcel pickup.

by jointly optimizing the control parameters of the UAV and UGV in an efficient manner.

The rest of this paper is organized as follows. Section II describes the system model for cooperative UAV–UGV delivery and formulates the corresponding optimization problem. Section III details the proposed two-stage iterative algorithm for UAV–UGV cooperation. Section IV presents simulation results that evaluate the performance of the proposed scheme against baseline methods under various system settings. Finally, Section V concludes the paper with a summary of key findings.

II. System Model and Problem Statement

We consider a cooperative parcel pickup scenario where one UAV and one UGV depart from a warehouse, pick up a total of K parcels, and return to the warehouse with the collected parcels, as depicted in Fig. 1. Let T_m denote the period required for vehicle m to complete parcel pickup, where $m \in \mathcal{M} = \{U, G\}$ is an indicator of the vehicle, i.e., U for UAV and G for UGV. Moreover, T_m is discretized into N time slots of different lengths $\delta_m[n]$, such as $T_m = \sum_{n=1}^{N} \delta_m[n]$ with $n \in \mathcal{N} = \{1, 2, \ldots, N\}$. At the beginning of time slot n, the horizontal and vertical coordinates of vehicle m are denoted as $\mathbf{q}_m[n] = (x_m[n], y_m[n])$ and $z_m[n]$, respectively. The UAV should fly at the allowed flight altitude between $H_{\min}^U$ and $H_{\max}^U$, except when landing to pick up a parcel, and to avoid collisions, the distance between the vehicles must be greater than the minimum distance, $d_{\min}$, in every time slot. We can then build the following constraints.

$$\|\mathbf{q}_m[n] - \mathbf{q}_{\text{str}}\|^2 + (z_m[n] - z_{\text{str}})^2 \le R_{\text{str}}^2, \ \forall m, n \in \{0, N\}, \tag{1}$$

$$\|\mathbf{q}_U[n] - \mathbf{q}_{\text{str}}\|^2 \le R_{\text{str}}^2, \ \forall n \in \{1, N-1\}, \tag{2}$$

$$\|\mathbf{q}_U[n] - \mathbf{q}_G[n]\|^2 + (z_U[n] - z_G[n])^2 \ge d_{\min}^2, \ \forall n, \tag{3}$$

where $(\mathbf{q}_{\text{str}}, z_{\text{str}})$ is the horizontal and vertical coordinates of the warehouse and R_{str} is the radius of the warehouse with $R_{\text{str}} \ge d_{\min}$. Note that constraint (2) prevents the UAV from deviating from the allowed flight altitude when taking off or landing from the warehouse.

It should be noted that, whereas the UAV can fly freely in 3D space, the UGV can only move either horizontally or vertically using elevators at any given moment. Accordingly, we can design the mobility constraints for both vehicles as follows: Let V_U and V_U^{v} denote the maximum flight velocity of the UAV in the 3D space and the vertical direction, respectively, where V_U^{v} is generally smaller than V_U [16]. Also, let V_G^{h} and V_G^{v} denote the maximum horizontal and vertical velocities of the UGV, respectively. Moreover, $s_G[n]$ is a UGV movement direction indicator, i.e., $s_G[n] = 1$ indicates that the UGV is moving in the horizontal direction, otherwise $s_G[n] = 0$. These maximum velocities limit the maximum allowable distance that each vehicle can travel in each time slot, which can be represented by

$$s_G[n] \in \{0, 1\}, \ \forall n, \tag{4}$$

$$\|\mathbf{q}_G[n] - \mathbf{q}_G[n-1]\| \le s_G[n] V_G^{\text{h}} \delta_G[n], \ \forall n, \tag{5}$$

$$\|z_G[n] - z_G[n-1]\| \le (1 - s_G[n]) V_G^{\text{v}} \delta_G[n], \ \forall n, \tag{6}$$

$$\|\mathbf{q}_U[n] - \mathbf{q}_U[n-1]\|^2 + (z_U[n] - z_U[n-1])^2 \le V_U^2 \delta_U^2[n], \ \forall n, \tag{7}$$

$$\|z_U[n] - z_U[n-1]\| \le V_U^{\text{v}} \delta_U[n], \ \forall n. \tag{8}$$

We also consider L nonoverlapping NFZs of cylindrical shape that the UAV is not allowed to access but the UGV is. Using the concept of an expanded NFZ introduced in [17], we can build the following additional constraint for the UAV to completely avoid the NFZs during a continuous trajectory.

$$\|\mathbf{q}_U[j] - \mathbf{c}_l\|^2 \ge R_l^2 + \left(\frac{V_U \delta_U[n]}{2}\right)^2, \ \forall l, n, j \in \{n-1, n\}, \tag{9}$$

where $\mathbf{c}_l = (x_l, y_l)$ and R_l are the coordinate center and radius of NFZ l with $l \in \mathcal{L} = \{1, 2, \cdots, L\}$.

To design the pickup of parcels, let $\alpha_{m,k}[n]$ denote a binary pickup indicator for parcel k by vehicle m at time slot n, i.e., $\alpha_{m,k}[n] = 1$ indicates that parcel k is picked up by vehicle m at time slot n, and $\alpha_{m,k}[n] = 0$ otherwise. Parcel k weighs w_k and vehicle m has a maximum load W_m that it can carry, so the total weight of the parcel being carried by vehicle m must be less than or equal to the maximum allowable weight for every time slot. Therefore, the following constraints can be constructed.

$$\alpha_{m,k}[n] \in \{0, 1\}, \ \forall m, k, n, \tag{10}$$

$$\sum_{m \in \mathcal{M}} \sum_{n=1}^{N} \alpha_{m,k}[n] \le 1, \ \forall k, \tag{11}$$

$$\sum_{k=1}^{K} \alpha_{m,k}[n] \le 1, \ \forall m, n, \tag{12}$$

$$w_m[n] = \sum_{k=1}^{K} \left(\sum_{j=1}^{n} \alpha_{m,k}[j] \right) w_k \le W_m, \ \forall m, n. \tag{13}$$

Here, constraint (11) implies that parcel k must be picked up by only one vehicle during the entire time slot and constraint (12) represents that vehicle m can pick up only one parcel at each time slot.

When picking up a parcel, the UAV can descend below $H_{\min}^U$, which is outside the allowed altitude range between $H_{\min}^U$ and $H_{\max}^U$. The UGV can also move vertically within $H_{\max}^G$ when picking up a parcel at a high altitude. These constraints can be expressed as

$$\left(1 - \sum_{k=1}^{K} \alpha_{U,k}[n]\right) H_{\min}^U \le z_U[n] \le H_{\max}^U, \ \forall n, \tag{14}$$

$$0 \le z_G[n] \le (1 - s_G[n]) \, H_{\max}^G, \ \forall n, \tag{15}$$

$$0 \le z_G[n] \le \left(\sum_{k=1}^{K} \alpha_{G,k}[n]\right) H_{\max}^G, \ \forall n. \tag{16}$$

Note that, for the UGV to move up, one of $\alpha_{G,k}[n]$ $\forall k$ is set to 1, and at the same time $s_G[n]$ is set to 0 at time slot n.

To pick up parcel k, each vehicle needs to have access to a circular pickup zone with a tolerance of $\mathcal{R}_k$. The UAV (UGV) must be within this pickup zone both when retrieving the parcel and while descending (ascending) to pick up the parcel and ascending (descending) after picking up the parcel. Consequently, the vehicles must be in the pickup zone during the time slot before and after the parcel is picked up; this can be formulated as follows:

$$\alpha_{m,k}[n](\|\mathbf{q}_m[n]-\mathbf{u}_k\|^2+(z_m[n]-z_k)^2) \le \mathcal{R}_k^2, \tag{17}$$

$$\alpha_{m,k}[n](\|\mathbf{q}_m[j]-\mathbf{u}_k\|^2) \le \mathcal{R}_k^2, \quad \forall m, k, n \neq N, j \in \{n-1, n+1\}, \tag{18}$$

where $\mathbf{u}_k = (x_k, y_k)$ and z_k are the horizontal center and vertical coordinates of the k-th pickup zone, respectively.

The constraint for indicating the pickup of all parcels, i.e., completion of the mission, is represented by

$$\sum_{m\in\mathcal{M}}\sum_{k=1}^{K}\sum_{n=1}^{N}\alpha_{m,k}[n] \ge K. \tag{19}$$

In this study, we aim to optimize the UGV movement direction indicator ($\mathbf{S} \triangleq \{s_G[n], \forall n\}$), pickup indicator ($\mathbf{A} \triangleq \{\alpha_{m,k}[n], \forall m, k, n\}$), horizontal and vertical trajectories of the vehicles ($\mathbf{Q} \triangleq \{\mathbf{q}_m[n], \forall m, n\}$ and $\mathbf{Z} \triangleq \{z_m[n], \forall m, n\}$), and lengths of the time slots ($\Delta \triangleq \{\delta_m[n], \forall m, n\}$) with the aim of minimizing pickup completion time while ensuring that there are no violations of NFZs. Defining $\eta = \max_{m\in\mathcal{M}}\left(\sum_{n=1}^{N}\delta_m[n]\right)$, we can formulate the optimization problem as follows:

$$\begin{aligned}
\textbf{(P0):}\quad & \min_{\mathbf{S},\,\mathbf{A},\,\mathbf{Q},\,\mathbf{Z},\,\Delta} \quad \eta \\
\text{s. t.}\quad & \eta \ge \sum_{n=1}^{N}\delta_m[n],\ \forall m, \\
& (1) - (19).
\end{aligned} \tag{20}$$

Problem **(P0)** is a nonconvex mixed-integer nonlinear programming (MINLP) because $\mathbf{S}$ and $\mathbf{A}$ are the sets of binary variables and constraints (3), (5), (6), (7), (9), (17), and (18) are not convex sets with respect to (w.r.t.) the relevant optimization variables. It is therefore difficult to mathematically derive a globally optimal solution.

III. Proposed Algorithm

To overcome the challenges stemming from the nonconvex constraints in problem **(P0)**, we use SCA to convert them into convex sets for optimization variables. This enables the use of well-established convex optimization solvers, such as CVX, which utilizes interior-point methods [18], to efficiently obtain solutions. We also apply a relaxation approach to the binary variables $\mathbf{S}$ and $\mathbf{A}$, thus allowing them to take continuous values, and we utilize PCCP to preserve the essential binary nature of these variables. Lastly, we introduce a two-stage iterative algorithm to solve the relaxed convex problem. The specific procedure we use to address each nonconvex constraint is detailed below.

To deal with the binary constraints (4) and (10), we transform $s_G[n]$ and $\alpha_{m,k}[n]$ into the equivalent forms, as follows:

$$0 \le s_G[n] \le 1,\ \forall n, \tag{21}$$

$$0 \le \alpha_{m,k}[n] \le 1,\ \forall m, k, n, \tag{22}$$

$$s_G[n]\,(1-s_G[n]) \le 0,\ \forall n, \tag{23}$$

$$\alpha_{m,k}[n]\left(1-\alpha_{m,k}[n]\right) \le 0,\ \forall m, k, n, \tag{24}$$

where constraints (23) and (24) guarantee that the binary nature of $s_G[n]$ and $\alpha_{m,k}[n]$ is preserved. However, it is difficult to address these constraints because they are nonconvex sets and have a tight search space. Based on the fact that the left-hand side (LHS) of (23) and (24) is concave w.r.t. $s_G[n]$ and $\alpha_{m,k}[n]$, respectively, and that a concave function is upper-bounded by its first-order Taylor expansion at any point [19], we can modify (23) and (24) by deriving their respective upper bounds, as follows:

$$(s_G^r[n])^2+s_G[n](1-2s_G^r[n]) \le \rho[n],\ \forall n, \tag{25}$$

$$(\alpha_{m,k}^r[n])^2+\alpha_{m,k}[n](1-2\alpha_{m,k}^r[n]) \le \lambda_{m,k}[n],\ \forall m, k, n. \tag{26}$$

Here, $\rho[n] \ge 0$ and $\lambda_{m,k}[n] \ge 0$ are nonnegative slack variables used for PCCP, which increase the initial feasible region for an efficient update to optimize $s_G[n]$ and $\alpha_{m,k}[n]$. Further, $s_G^r[n]$ and $\alpha_{m,k}^r[n]$ are the values of $s_G[n]$ and $\alpha_{m,k}[n]$ updated for the r-th iteration.

To address the nonconvexity of (3), we use the first-order Taylor expansion to derive the lower bound on the LHS of (3) because it is convex w.r.t. the optimization variables. Then, constraint (3) can be converted into the following convex set.

$$\begin{aligned}
&\|\mathbf{q}_U[n]-\mathbf{q}_G[n]\|^2+(z_U[n]-z_G[n])^2 \ge -\|\mathbf{q}_U^r[n]-\mathbf{q}_G^r[n]\|^2 \\
&\quad + 2\left(\mathbf{q}_U^r[n]-\mathbf{q}_G^r[n]\right)^T(\mathbf{q}_U[n]-\mathbf{q}_G[n]) - (z_U^r[n]-z_G^r[n])^2 \\
&\quad + 2\left(z_U^r[n]-z_G^r[n]\right)(z_U[n]-z_G[n]) \ge d_{\min}^2,\ \forall n,
\end{aligned} \tag{27}$$

where $\mathbf{q}_m^r[n]$ and $z_m^r[n]$ are the values of $\mathbf{q}_m[n]$ and $z_m[n]$ updated for the r-th iteration with $m \in \{U, G\}$.

Given that $s_G[n]\delta_G[n]$ can be replaced with $\frac{(s_G[n]+\delta_G[n])^2}{2} - \frac{s_G^2[n]+\delta_G^2[n]}{2}$, we can convert (5) into the equivalent form, as follows:

$$\begin{aligned}
&\|\mathbf{q}_G[n]-\mathbf{q}_G[n-1]\| + V_G^{\mathrm{h}}\left(\frac{s_G^2[n]+\delta_G^2[n]}{2}\right) \\
&\quad \le V_G^{\mathrm{h}}\left(\frac{(s_G[n]+\delta_G[n])^2}{2}\right),\ \forall n.
\end{aligned} \tag{28}$$

Applying the first-order Taylor expansion to the right-hand side (RHS) of (28), we can make (28) a convex set as follows:

$$\begin{aligned}
&\|\mathbf{q}_G[n]-\mathbf{q}_G[n-1]\| + V_G^{\mathrm{h}}\left(\frac{s_G^2[n]+\delta_G^2[n]}{2}\right) \le V_G^{\mathrm{h}} \\
&\times\left(-\frac{(s_G^r[n]+\delta_G^r[n])^2}{2} + (s_G^r[n]+\delta_G^r[n])(s_G[n]+\delta_G[n])\right),\ \forall n,
\end{aligned} \tag{29}$$

where $\delta_G^r[n]$ is the value of $\delta_G[n]$ updated for the r-th iteration.

Using a similar approach, (6) and (7) can be transformed into the following convex sets.

$$\begin{aligned}
&\|z_G[n]-z_G[n-1]\| + V_G^{\mathrm{v}}\left(\frac{(s_G[n]+\delta_G[n])^2}{2}\right) - V_G^{\mathrm{v}}\delta_G[n] \le V_G^{\mathrm{v}} \\
&\times\left(-\frac{(s_G^r[n])^2+(\delta_G^r[n])^2}{2} + s_G^r[n]s_G[n] + \delta_G^r[n]\delta_G[n]\right),\ \forall n,
\end{aligned} \tag{30}$$

$$\begin{aligned}
&\|\mathbf{q}_U[n]-\mathbf{q}_U[n-1]\|^2+(z_U[n]-z_U[n-1])^2 \\
&\quad \le V_U^2\left((\delta_U^r[n])^2 + 2\delta_U^r[n]\left(\delta_U[n]-\delta_U^r[n]\right)\right),\ \forall n,
\end{aligned} \tag{31}$$

where $\delta_U^r[n]$ is the value of $\delta_U[n]$ updated for the r-th iteration.

The LHS of (9) is convex w.r.t $\mathbf{q}_U[j]$, so we can apply the first-order Taylor expansion to it to get a lower bound. Using this lower bound, we can replace constraint (9) with the following convex set.

$$\begin{aligned}
&\|\mathbf{q}_U^r[j]-\mathbf{c}_l\|^2 + 2(\mathbf{q}_U^r[j]-\mathbf{c}_l)^T(\mathbf{q}_U[j]-\mathbf{q}_U^r[j]) \\
&\quad \ge R_l^2 + \left(\frac{V_U\delta_U[n]}{2}\right)^2,\ \forall l, n, j \in \{n-1, n\}.
\end{aligned} \tag{32}$$

To handle the nonconvexity of (17), we convert it to the following equivalent form.

$$\alpha_{m,k}[n] \leq \frac{\mathcal{R}_k^2+\varepsilon^2}{\|\mathbf{q}_m[n]-\mathbf{u}_k\|^2+(z_m[n]-z_k)^2+\varepsilon^2}, \quad (33)$$

where ε is a positive constant that prevents division by zero. Note that (33) is not still a convex set, but that its RHS is convex w.r.t. $\|\mathbf{q}_m[n]-\mathbf{u}_k\|^2+(z_m[n]-z_k)^2$. Therefore, we can make (33) a convex set by deriving its lower bound, as follows:

$$\alpha_{m,k}[n] \leq \frac{2(\mathcal{R}_k^2+\varepsilon^2)}{\|\mathbf{q}_m^r[n]-\mathbf{u}_k\|^2+(z_m^r[n]-z_k)^2+\varepsilon^2} - \frac{(\mathcal{R}_k^2+\varepsilon^2)\left(\|\mathbf{q}_m[n]-\mathbf{u}_k\|^2+(z_m[n]-z_k)^2+\varepsilon^2\right)}{\left(\|\mathbf{q}_m^r[n]-\mathbf{u}_k\|^2+(z_m^r[n]-z_k)^2+\varepsilon^2\right)^2}. \quad (34)$$

Similarly, (18) can also be transformed to the following convex set.

$$\alpha_{m,k}[n] \leq \frac{2(\mathcal{R}_k^2+\varepsilon^2)}{\|\mathbf{q}_m^r[j]-\mathbf{u}_k\|^2+\varepsilon^2} - \frac{(\mathcal{R}_k^2+\varepsilon^2)\left(\|\mathbf{q}_m[j]-\mathbf{u}_k\|^2+\varepsilon^2\right)}{\left(\|\mathbf{q}_m^r[j]-\mathbf{u}_k\|^2+\varepsilon^2\right)^2}, \quad \forall m,k,n\neq N,\ j\in\{n-1,n+1\}. \quad (35)$$

With the modified convex constraints, problem **(P0)** can be converted into the following convex optimization problem:

$$\textbf{(P1):} \min_{\mathbf{S},\,\mathbf{A},\,\mathbf{Q},\,\mathbf{Z},\,\Delta,\,\boldsymbol{\rho},\,\boldsymbol{\lambda}} \quad \eta + \mu\mathcal{P}(\boldsymbol{\rho}) + \kappa\mathcal{P}(\boldsymbol{\lambda})$$
$$\text{s. t.} \quad (1),\ (2),\ (8),\ (11)-(16),\ (19)-(22),\ (25)-(27),\ (29)-(32),\ (34),\ (35),$$

where $\boldsymbol{\rho} \triangleq \{\rho[n] \geq 0, \forall n\}$, $\boldsymbol{\lambda} \triangleq \{\lambda_{m,k}[n] \geq 0, \forall m,k,n\}$, $\mathcal{P}(\boldsymbol{\rho}) = \sum_{n=1}^{N} \rho[n]$, and $\mathcal{P}(\boldsymbol{\lambda}) = \sum_{m\in\mathcal{M}} \sum_{k=1}^{K} \sum_{n=1}^{N} \lambda_{m,k}[n]$. Note that $\mathcal{P}(\boldsymbol{\rho})$ and $\mathcal{P}(\boldsymbol{\lambda})$ are the penalty terms to enforce $\mathbf{S}$ and $\mathbf{A}$ to be binary, while $\mu > 0$ and $\kappa > 0$ are the variables that control the impact of the penalty terms on the objective function. Initially, for small values of μ and κ, the optimization variables are optimized to minimize pickup completion time η, thus allowing for some flexibility for $\mathbf{S}$ and $\mathbf{A}$ such that they do not strictly adhere to the binary property. However, with increasing values of μ and κ, the optimization variables are optimized to minimize η while ensuring that $\mathbf{S}$ and $\mathbf{A}$ remain binary so that the penalty terms are zero [20].

In problem **(P1)**, the pickup completion time for each vehicle is optimized to the same value because η is defined as $\eta = \max_{m\in\mathcal{M}} \left(\sum_{n=1}^{N} \delta_m[n]\right)$. This means that the vehicle that can complete pickup earlier has to wait unnecessarily for another vehicle to complete pickup, which can result in a wasted trajectory during this time. To prevent this from happening, we further optimize $\mathbf{Q}$ and $\mathbf{Z}$ to minimize the sum of the distances traveled by the vehicles, which is defined as $\Upsilon = \sum_{m\in\mathcal{M}} \sum_{n=1}^{N} \sqrt{\|\mathbf{q}_m[n]-\mathbf{q}_m[n-1]\|^2 + (z_m[n]-z_m[n-1])^2}$, for the fixed values of the other variables optimized in **(P1)**, as follows:

$$\textbf{(P2):} \min_{\mathbf{Q},\,\mathbf{Z}} \quad \Upsilon$$
$$\text{s. t.} \quad (1),\ (2),\ (8),\ (14)-(16),\ (27),\ (29)-(32),\ (34),\ (35).$$

Note that in problem **(P2)**, Δ that has been optimized in **(P1)** is fixed, so the pickup completion time does not change, and only the trajectory of each vehicle is efficiently optimized.

Algorithm 1 presents the overall procedure of the proposed iterative algorithm. The algorithm begins with a set of initial feasible values, including all optimization variables and associated penalty control parameters. In each iteration, the penalty parameters are scaled by a constant factor greater than one, continuing until they reach their predefined maximum limits. As established in Theorem 1 of [21], a finite upper bound exists for each penalty parameter, ensuring that the penalty function converges to zero. Once this condition is met, the solution is updated iteratively in a decreasing direction. In other words, from this point on, the objective function values in problems **(P1)** and **(P2)** exhibit nonincreasing behavior with each iteration, and given that the objective function is also bounded by a finite value [22], the convergence of the proposed algorithm is guaranteed.

Algorithm 1 Proposed Algorithm

1: Set $r = 0$
2: Initialize $\mathbf{S}^r$, $\mathbf{A}^r$, $\mathbf{Q}^r$, $\mathbf{Z}^r$, Δ^r, $\boldsymbol{\rho}^r$, $\boldsymbol{\lambda}^r$, μ^r, κ^r, ε^r, $\mu_{\max}$, $\kappa_{\max}$, $\varepsilon_{\min}$, $\sigma_\mu > 1$, $\sigma_\kappa > 1$, and $\sigma_\varepsilon < 1$
3: Calculate $f^r = \eta^r + \mu^r\mathcal{P}(\boldsymbol{\rho}^r) + \kappa^r\mathcal{P}(\boldsymbol{\lambda}^r)$
4: **repeat**
5: $\quad r \leftarrow r+1$
6: $\quad f^{\text{old}} \leftarrow f^{r-1}$
7: $\quad$ Find $\{\mathbf{S}^r, \mathbf{A}^r, \mathbf{Q}^r, \mathbf{Z}^r, \Delta^r, \boldsymbol{\rho}^r, \boldsymbol{\lambda}^r\}$ by solving **(P1)** for given $\{\mathbf{S}^{r-1}, \mathbf{A}^{r-1}, \mathbf{Q}^{r-1}, \mathbf{Z}^{r-1}, \Delta^{r-1}\}$
8: $\quad$ Update $\mu^r = \min\{\sigma_\mu\mu^{r-1}, \mu_{\max}\}$, $\kappa^r = \min\{\sigma_\kappa\kappa^{r-1}, \kappa_{\max}\}$, and $\varepsilon^r = \max\{\sigma_\varepsilon\varepsilon^{r-1}, \varepsilon_{\min}\}$
9: $\quad$ Calculate $f^r = \eta^r + \mu^r\mathcal{P}(\boldsymbol{\rho}^r) + \kappa^r\mathcal{P}(\boldsymbol{\lambda}^r)$
10: **until** $|f^r - f^{\text{old}}| \leq \epsilon_1$
11: Calculate $f^r = \Upsilon^r$
12: **repeat**
13: $\quad r \leftarrow r+1$
14: $\quad f^{\text{old}} \leftarrow f^{r-1}$
15: $\quad$ Find $\{\mathbf{Q}^r, \mathbf{Z}^r\}$ by solving **(P2)** for given $\{\mathbf{Q}^{r-1}, \mathbf{Z}^{r-1}\}$
16: $\quad$ Update $\varepsilon^r = \max\{\sigma_\varepsilon\varepsilon^{r-1}, \varepsilon_{\min}\}$
17: $\quad$ Calculate $f^r = \Upsilon^r$
18: **until** $|f^r - f^{\text{old}}| \leq \epsilon_2$

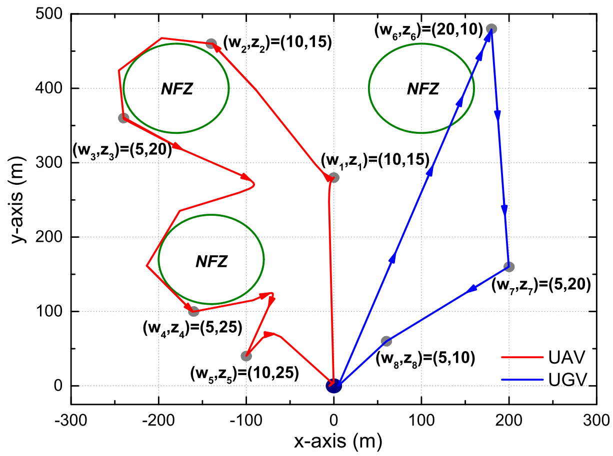
The computational complexity is also expressed as $O\left(\max\left[(MKN)^{3.5}R_1\log(1/\epsilon_1), (MN)^{3.5}R_2\log(1/\epsilon_2)\right]\right)$, where R_1 and R_2 imply the number of iterations that is required to converge the first loop (lines 4–10) and the second loop (lines 12–18), respectively. Given that the first loop is dominant in determining the computational complexity of the proposed algorithm, it can be noted that the proposed algorithm exhibits a polynomial time computational complexity of MKN, making it suitable for real-time implementation [23]. On the other hand, commercial solvers, such as Gurobi and BARON, have an exponential computational complexity of $O(2^{MKN})$, making them infeasible to solve problems in real-time.

IV. Simulation Results and Discussions

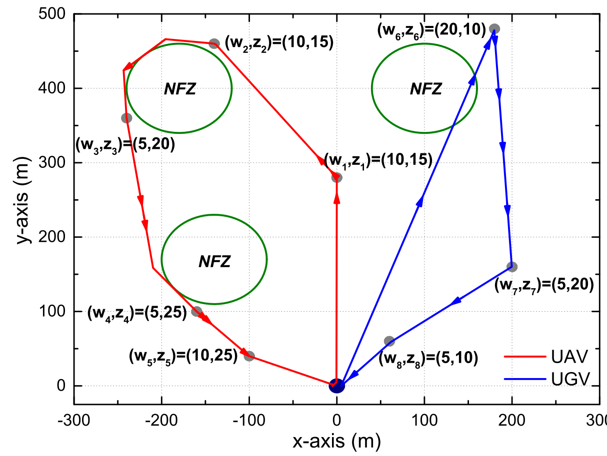
For performance evaluations, we consider the following parameters as default values [6], [16], [17]: $M = 2$, $N = 100$, $K = 8$, $L = 3$, $H_{\min}^U = 50$ m, $H_{\max}^U = 100$ m, $H_{\max}^G = 50$ m, $V_G^{\text{h}} = 5$ m/s, $V_G^{\text{v}} = 2$ m/s, $V_U = 10$ m/s, $V_U^{\text{v}} = 5$ m/s, $R_l = 60$ m, $W_G = W_U = 70$ kg, $\mathcal{R}_k = 2$ m, $\epsilon_1 = \epsilon_2 = 10^{-3}$, $\mu^0 = 0.15$, $\kappa^0 = 0.7$, $\varepsilon^0 = 50$, $\sigma_\mu = 1.12$, $\sigma_\kappa = 1.4$, $\sigma_\varepsilon = 0.9$, $\mu_{\max} = 10^6$, $\kappa_{\max} = 10^6$, and $\varepsilon_{\min} = 19$. Further, the coordinates of the warehouse are fixed as $(\mathbf{q}_{\text{str}}, z_{\text{str}}) = (0,0,0)$ m, while pickup zones and NFZs are randomly deployed within an area having a size of 500 m $\times$ 500 m.

For performance comparisons, we consider the following five schemes:

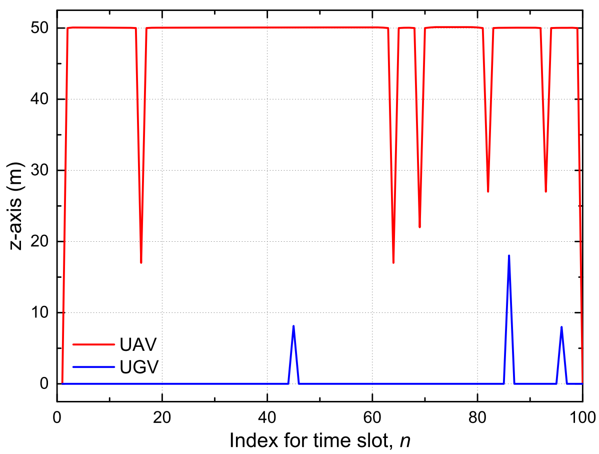
1) *Proposed scheme*: The trajectory and pickup strategy of the UAV and UGV are determined by Algorithm 1.



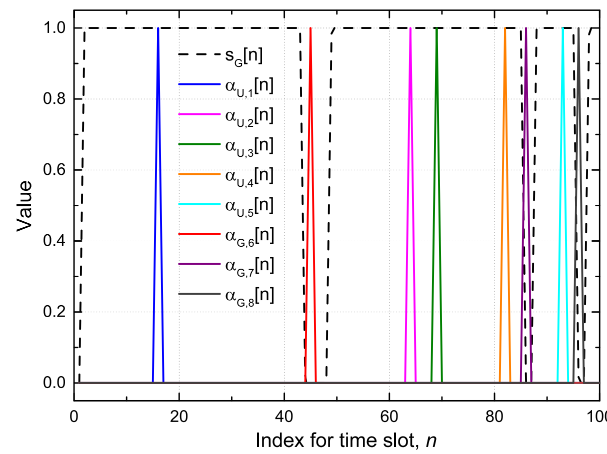
(a) Horizontal trajectory solved for **(P1)**.



(b) Final horizontal trajectory solved for **(P1)** and **(P2)**.



(c) Final vertical trajectory.

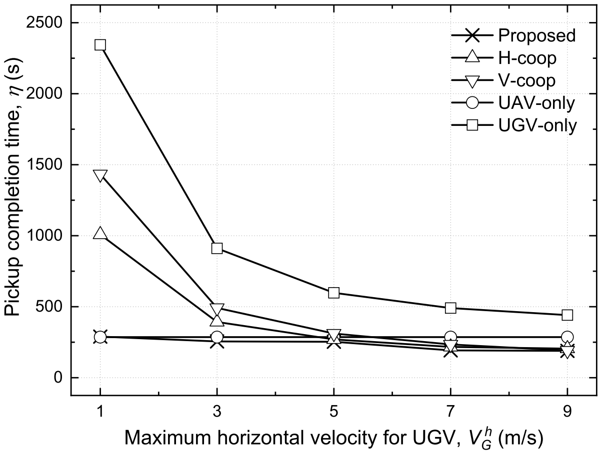


(d) Binary indicators for pickup strategy.

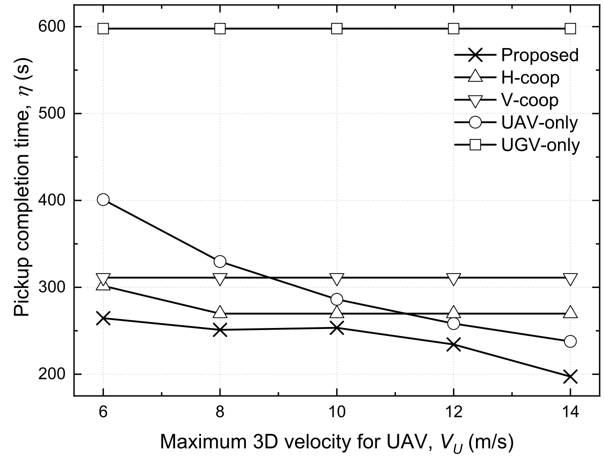
Fig. 2. Trajectory and resource allocation of the proposed scheme.

2) *Horizontal cooperative (H-coop) scheme*: $\mathbf{A}$ is fixed such that the UAV preferentially picks up parcels that are located farther away from the warehouse and the UGV picks up parcels that are located closer to the warehouse, while the remaining variables, $\mathbf{S}$, $\mathbf{Q}$, $\mathbf{Z}$, and Δ, are determined by Algorithm 1.
3) *Vertical cooperative (V-coop) scheme*: $\mathbf{A}$ is fixed such that the UAV preferentially picks up parcels that are located at higher elevations and the UGV picks up parcels that are located at lower elevations, while the remaining variables, $\mathbf{S}$, $\mathbf{Q}$, $\mathbf{Z}$, and Δ, are determined by Algorithm 1.
4) *UAV-only scheme*: The trajectory and pickup strategy of the UAV are determined by Algorithm 1 without the UGV.
5) *UGV-only scheme*: The trajectory and pickup strategy of the UGV are determined by Algorithm 1 without the UAV.

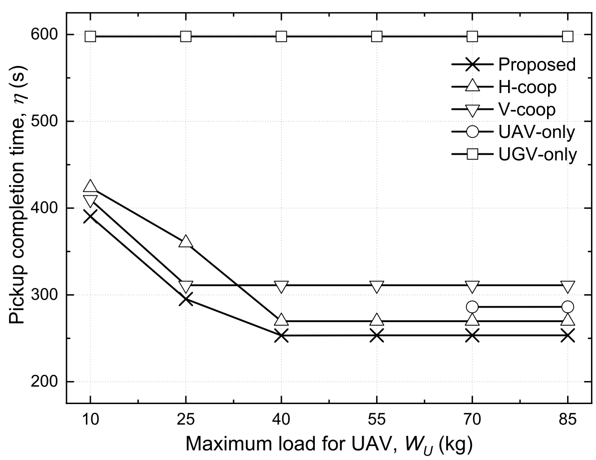
Fig. 2 shows the trajectory and resource allocation of the proposed scheme: (a) horizontal trajectory solved for **(P1)**, (b) final horizontal trajectory solved for **(P1)** and **(P2)**, (c) final vertical trajectory, and (d) binary indicators for pickup strategy. In Figs. 2(a) and 2(b), the pickup zones are marked with grey circles that show the weight of the parcel and the height of the pickup zone, while the warehouse is marked with a blue circle. The objective of problem **(P1)** is to minimize the maximum pickup completion times of both the UAV and the UGV, so the control variables are optimized such that the pickup completion times of the UAV and the UGV are as close to equal to each other as possible. In this case, the slower UGV takes the shortest route to reduce the completion time, but the faster UAV can complete the pickup in the same amount of time without taking the shortest route. As a result, as shown in Fig. 2(a), the UAV takes an unnecessary route because the objective in problem **(P1)** does not consider the travel distance of each vehicle. As shown in Fig. 2(b), by optimizing $\mathbf{Q}$ and $\mathbf{Z}$ in problem **(P2)** for the fixed $\mathbf{S}$, $\mathbf{A}$, and Δ optimized in problem **(P1)**, we can further optimize the trajectory of the UAV with little change in the distance traveled by the UGV which is already optimized for the shortest path. Through this additional optimization, we can reduce the total distance traveled by the UAV and UGV from 3004 m to 2669 m. We can also observe that the UAV smoothly avoids each NFZ without violating it, while the UGV can pass through NFZs. Fig. 2(c) confirms that in each pickup zone, the UAV (UGV) descends (ascends) in the z-axis direction to pick up the parcel within a tolerance of $\mathcal{R}_k = 2$ m. Fig. 2(d) also shows that the UGV movement direction indicator ($s_G[n]$) and pickup indicator ($\alpha_{m,k}[n]$) retain their binary nature in the proposed algorithm. For example, the pickup indicator of the UAV is set to 1 when it picks up the parcel and set to 0 otherwise. Moreover, when the UGV movement direction indicator is set to 0, the pickup indicator of the UGV is



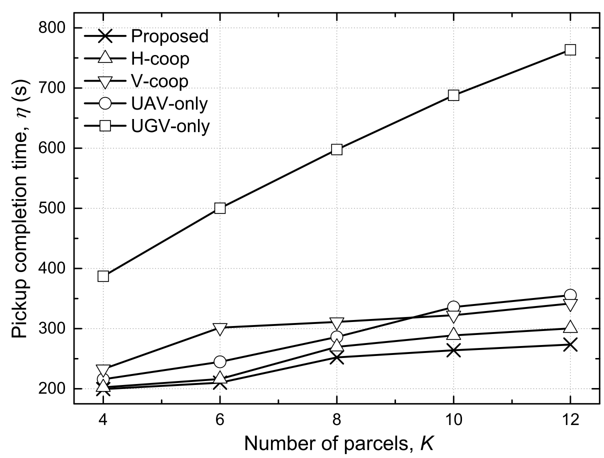
(a) η vs. V_G^{h}.



(b) η vs. V_U.



(c) η vs. W_U.



(d) η vs. K.

Fig. 3. Performance comparison.

also set to 1, indicating that it will move up only to pick up the parcel.

Fig. 3 shows the performance comparison in terms of pickup completion time (η) between the proposed and baseline schemes for important parameters: (a) maximum horizontal velocity for UGV (V_G^{h}), (b) maximum 3D velocity for UAV (V_U), (c) maximum load for UAV (W_U), and (d) number of parcels (K).

As shown in Fig. 3(a), the η of the UAV-only scheme does not change even as V_G^{h} increases, because its pickup completion time is not affected by V_G^{h}. On the other hand, the η of the other schemes significantly improves with increasing V_G^{h}. This is because the velocity of the UGV is slower than that of the UAV, making V_G^{h} the dominant factor in determining the final pickup completion time in these schemes. For the same reason, the UGV-only scheme shows the lowest performance. Moreover, the H-coop scheme achieves shorter η than the V-coop scheme because V_G^{h} enhances the horizontal movement of the UGV. When V_G^{h} is small, the H-coop and V-coop schemes are both outperformed by the UAV-only scheme, because the pickup indicator is fixed in the two cooperative schemes and the slow velocity of the UGV determines their final pickup completion times. However, with increasing V_G^{h}, the H-coop and V-coop schemes outperform the UAV-only scheme due to the increased gains from cooperation. The proposed scheme accomplishes the shortest η by jointly optimizing all control parameters to enable load balancing for effective cooperation.

In Fig. 3(b), the UGV-only scheme does not depend on V_U, so η remains constant regardless of V_U. In the H-coop scheme, the UAV needs more time than the UGV to complete the pickup when $V_U =$ 6 m/s due to its slow velocity. However, as V_U increases above 8 m/s, the pickup completion time of the UAV improves, and the final performance is determined by the time the UGV completes the pickup independent of V_U, resulting in a constant η for $V_U \geq 8$ m/s. For the same reason, the η of the V-coop scheme is constant for all ranges of V_U, because the UGV always requires more time than the UAV to complete the pickup. On the other hand, the η of the UAV-only scheme improves considerably as V_U increases, which is attributed to the enhanced movement of the UAV. As a result, it can achieve better performance than the H-coop and V-coop schemes when $V_U \geq 12$ m/s. Comparing Figs. 3(a) and 3(b), we can see that the performance improvement with increasing V_G^{h} is greater than that with increasing V_U, and we can infer that the velocity of the UGV is a more important factor in determining the pickup completion time than that of the UAV.

In Fig. 3(c), the η of the UAV-only scheme is depicted only for $W_U \geq 70$ kg because the UAV cannot complete the pickup of all parcels when W_U is less than the total weight of the parcels, which is 70 kg. Further, independent of W_U, the UGV-only scheme achieves a constant η. For the remaining three cooperative schemes, the UGV should pick up more parcels when W_U is small, which leads to a notable increase in η, especially when $W_U \leq 40$ kg. However, as W_U increases above 40 kg, the UAV and UGV can split the parcels appropriately to pick them up, and this cooperative strategy does not change, which ultimately results in a constant performance.

In Fig. 3(d), the η of all schemes increases as K increases because the UAV and UGV need more time to pick up the increased number of parcels. When K is small, e.g., $K \leq 8$, the UAV-only scheme outperforms the V-coop scheme due to its fast maneuverability. However, as the number of parcels increases, the cooperative gain between the UAV and UGV increases, resulting in a performance reversal between the two schemes. Overall performance trends are similar to the other results. For Figs. 3(a)-3(d), the proposed scheme outperforms all the baseline schemes in terms of pickup completion time, which validates the scalability and robustness of the proposed method for practical applicability.

V. Conclusion

In this paper, we explored the joint optimization of 3D trajectories and pickup strategies for cooperative parcel pickup between UAV and UGV. Taking into account the movement characteristics of each vehicle and the avoidance of NFZs, we optimized the UGV movement direction indicator, pickup indicator, 3D trajectory, and lengths of time slots to identify the shortest route to minimize pickup completion time. To address the nonconvexity of the formulated problem, we leveraged SCA to transform it into a tractable convex problem and proposed a two-stage iterative algorithm using interior-point methods to find suboptimal solutions. We also used PCCP to maintain the binary characteristics of UGV movement direction and pickup indicators. By conducting simulations under various scenarios, we verified that the proposed scheme not only improves the pickup completion time compared to the baseline schemes but also efficiently forms the shortest route. We hope that our findings will be useful in designing practical cooperative strategies between UAV and UGV for parcel delivery.

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